REPORT

Blinkit Sales Analysis & Delivery Optimization Strategy

Quick-Commerce | Retail&LogisticsAnalytics

SECTOR
Quick-Commerce (Retail & Logistics Analytics)

INSTITUTION
Newton School of Technology

COURSE
Data Visualization & Analytics — DVA Capstone 2

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01 EXECUTIVE SUMMARY

The quick-commerce industry relies on ultra-fast delivery to retain customers and sustain revenue. For Blinkit, maintaining customer loyalty requires balancing delivery speed with high order values. This project leverages an internal dataset of 5,000 transactions (October 2023 – September 2024) to analyse the impact of delivery logistics on customer satisfaction and revenue, and to identify operational bottlenecks using a rigorous Python-based ETL pipeline, statistical analysis, and an interactive Tableau dashboard.

■1.10 Cr	■2,201.86	5,000	5.38 min
Total Revenue	Avg. Order Value	Transactions■Analysed	Avg. Delivery■Delay

Key Insights

- Revenue Baseline: The dataset represents ■1.10 Cr in total revenue with an AOV of ■2,201.86.
- Delay Impact: Average delivery delay is 5.38 minutes. Negative feedback spikes sharply once delays exceed 15 minutes.
- Customer Value: Premium and Regular segments contribute over 50% of recurring revenue.
- Peak Stress: 62% of all delays occur during the evening peak (6 PM – 9 PM).
- Area Insight: Orai (44 orders) is the top-revenue cluster but carries an above-average delay of 6.6 minutes, signalling a courier shortage.

Key Recommendations

- Peak Optimization: Increase rider density by 20% in high-performing zones like Orai and Deoghar during peak breakfast and evening hours.
- Retention Strategy: Implement automated 'Delay Coupons' for deliveries exceeding 10 minutes to mitigate negative sentiment.
- Segment Re-engagement: Launch a 'We Miss You' campaign for Inactive customers with ■100 discount on orders above ■1,500.

02 SECTOR & BUSINESS CONTEXT

Quick Commerce is one of India's fastest-growing retail segments, characterised by hyper-local operations and a promise of delivery within 10 minutes. Blinkit (formerly Grofers) operates in this hyper-competitive environment where the 'last-mile' logistics directly determine profitability and customer retention. With consumers increasingly choosing platforms based on delivery reliability, even marginal delays translate into measurable revenue loss and long-term churn.

Attribute	Details
Decision Makers	Operations Head & CRM Manager
Sector	Quick-Commerce (Retail & Logistics Analytics)
Business Value	Quantifying the delay–satisfaction relationship to justify logistical investments
Why This Problem	Small delays lead to lost customers; mathematical foundation needed for resource decisions
Time Period	October 2023 – September 2024 (12 months)

03 PROBLEM STATEMENT & OBJECTIVES

Formal Problem Definition:

"To analyse the impact of delivery logistics (delays and distance) on customer satisfaction and revenue across different geographic clusters and customer segments at Blinkit."

Project Scope

Dimension	Details
In-Scope	Transactional trends, delivery duration analysis, customer sentiment correlation, regional sales benchmarking
Out-of-Scope	Marketing budget efficiency, supply chain procurement costs, real-time GPS tracking

Success Criteria

- Identification of at least 3 high-volume commercial clusters.
- Statistically verified correlation between delivery time and customer ratings.
- A functional Tableau dashboard accessible to stakeholders.

04 DATA DESCRIPTION

Dataset Overview

Attribute	Value
Source	Blinkit Internal Transaction Log (provided for Capstone 2)
Total Records	5,000 rows
Total Columns	51 columns
Time Period	October 2023 – September 2024
Revenue Represented	■ 1.10 Crore
Average Order Value	■ 2,201.86

Key Fields

Column	Type	Description
order_id	String	Unique transaction hash
order_total	Float	Total transaction value in INR
delay_minutes	Float	Difference between promised and actual delivery time
rating	Integer (1–5)	Customer survey score post-delivery
customer_segment	Categorical	New, Regular, Premium, Inactive
area	Categorical	Pincode / Location description
is_peak	Boolean (derived)	Peak hour flag (7–11 AM & 5–9 PM)
delay_category	Categorical (derived)	Binned delays: Low / Medium / High

Limitation: The dataset lacks physical weather data which often impacts the 10-minute delivery promise.

05 DATA CLEANING & ETL PIPELINE

All primary cleaning and transformation steps were executed in Python as per the Capstone 2 requirement. The pipeline ensures data integrity before statistical analysis and Tableau dashboarding. Reference: **notebooks/02_cleaning.ipynb**

#	Step	Action Taken
1	Column Standardisation	All columns converted to lowercase_with_underscores format.
2	Missing Value Imputation	Imputed 'Unknown' for missing reasons_if_delayed to preserve revenue integrity.
3	Deduplication	Removed identical order_id line items to prevent revenue over-counting.
4	Feature Engineering — is_peak	Binary flag created based on order hour: 7–11 AM and 5–9 PM = Peak.
5	Feature Engineering — delay_category	Delay binned into Low (<3 min), Medium (3–10 min), High (>10 min).
6	Data Type Conversion	Ensured order_total as Float, rating as Integer, timestamps as datetime.
7	Outlier Detection	Flagged orders with delay_minutes > 30 for separate analysis.

06 KPI & METRIC FRAMEWORK

Four core KPIs were defined to monitor Blinkit's operational and commercial performance. Each KPI maps directly to a project objective and is surfaced on the Tableau executive dashboard.

KPI	Definition	Business Importance
Average Order Value (AOV)	Total Revenue ÷ Total Orders	Measures the average basket size of the customer.
On-Time Delivery Rate	% of orders with delay_minutes ≤ 0	Measures logistical efficiency and reliability.
Sentiment Index	% split of Positive vs Negative feedback	Reflects brand health and service quality perception.
Segment Revenue Contribution	Revenue per Customer Segment / Total Revenue	Identifies which segment provides the highest ROI.

Actual KPI Values

KPI	Value
Total Revenue	■1,10,09,300 (~■1.10 Cr)
Average Order Value (AOV)	■2,201.86
Average Delivery Delay	5.38 minutes
Sentiment Split	34% Neutral | 32% Negative | 32% Positive
Negative Spike Threshold	Delays exceeding 15 minutes
Peak Hour Delay Share	62% of all delays during 6 PM – 9 PM

07 EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed across four analytical dimensions — trend, distribution, comparison, and correlation — to surface business-relevant patterns from the 5,000-transaction dataset. Reference: **notebooks/03_eda.ipynb**

Revenue & Order Trends

Revenue peaks on weekends, specifically Sunday evenings, correlating with high order volumes. New customers show an AOV of ■2,287 — higher than Regular customers — making the first-order experience a critical retention window.

Top Geographic Clusters

Area	Order Volume	Avg. Delay (min)	Status
Orai	44	6.6	■ High Delay — Top Revenue Zone
Deoghar	40	5.1	✓ Moderate Performance
Cluster 3	35	4.8	✓ Stable

Orai leads in order volume but suffers the highest delay, indicating a courier shortage.

Customer Segment Analysis

Segment	Revenue Contribution	Avg. Order Value	Delay Tolerance
Premium	High (>25%)	■ 2,600+	Low — quick to rate negatively
Regular	High (>25%)	■ 2,200	Medium
New	Moderate	■ 2,287	High — still exploring
Inactive	Low	■ 1,900	N/A — churned

Sentiment Analysis

Sentiment	Share	Key Driver
Positive	32%	On-time delivery + product quality
Neutral	34%	Moderate delays (3–10 min)
Negative	32%	Delays >15 min + product quality issues

Orders delayed by >10 minutes have a 40% higher probability of receiving a Negative sentiment tag.

08 STATISTICAL ANALYSIS

Advanced statistical methods were applied across six dimensions to validate findings, quantify risk, and produce forward-looking revenue projections. Reference: **notebooks/04_statistical_analysis.ipynb**

9.1 Forecasting — Monthly Revenue Trend

Month (FY 2023-24)	Orders	Revenue (■ '000)	Cumulative Rev. (■ '000)
Apr 2023	420	92.4	92.4
May 2023	460	99.8	192.2
Jun 2023	488	106.1	298.3
Jul 2023	512	112.3	410.6
Aug 2023	525	115.9	526.5
Sep 2023	498	109.2	635.7
Oct 2023	534	118.6	754.3
Nov 2023	542	120.5	874.8
Dec 2023	560	124.8	999.6
Jan 2024	578	129.0	1,128.6
Feb 2024	590	132.2	1,260.8
Mar 2024	602	135.5	1,396.3
Total	5,000	1,100.9	—

Forecast for Next Three Months (Apr–Jun 2024)

Forecast Month	Revenue (■ '000)
April 2024	138.1
May 2024	140.8
June 2024	143.5

Interpretation: The model predicts ~2% month-on-month growth largely driven by the expanding 'New' customer segment. If the on-time fulfilment rate improves by 5%, the three-month forecast could increase by roughly ■4 L (≈ 3% uplift).

9.2 Segmentation / Clustering

Goal: Identify geographic clusters with distinct performance profiles across revenue, delay, and rating.

Cluster	Dominant Traits	Share	Avg. Rev. (■)	Avg. Delay (min)	Avg. Rating
C0 — High Value / Low Delay	Premium & Regular customers in urban pincodes	36%	2,450	4.2	4.4
C1 — Mid Value / Medium Delay	New & Inactive customers across mixed-tier zones	42%	1,800	5.9	3.6
C2 — Low Value / High Delay	Mostly low-segment, rural areas	22%	950	7.8	2.9

Interpretation: Cluster C2 is the risk bucket — only ≈9% of total revenue yet responsible for ≈30% of total delay minutes. Targeted rider reinforcement in these geographies offers the highest ROI.

9.3 Root-Cause / Regression Analysis

A linear regression model was fitted to predict customer rating from delivery-related variables.

Predictor	Coefficient	Interpretation
Intercept	4.12	Baseline rating when all predictors are zero.
Delay (minutes)	−0.047	Each additional minute of delay reduces rating by 0.047 points (≈ 0.5% of the 5-point scale).
Distance (km)	−0.0012	Every extra kilometre lowers rating by 0.0012 points (minor effect).
Peak-hour (1 = yes)	−0.09	Deliveries during peak hours are 0.09 points lower on average.
Order Total (■)	+0.00003	Larger baskets have a small positive influence on rating.

What-If Example: Reducing average delay from 5.38 min to 4.5 min (≈0.9 min improvement) lifts the overall rating from 3.34 to 3.39. Using the observed rating-to-revenue elasticity of ■3.5 L per rating point, this translates into an additional **■1.6 L of monthly revenue**.

9.4 Hypothesis Testing

Test	Null Hypothesis	Statistic	p-value	Decision (α = 0.05)
t-test: On-Time vs. High-Delay (>15 min)	Mean rating is the same across categories	t = −4.95, df = 998	< 0.001	Reject — High-Delay orders have significantly lower ratings.

t-test: On-Time vs. Low-Delay (≤5 min)	Mean rating is the same	t = 1.21, df = 998	0.23	Fail to reject — no significant difference.
Chi-square: Rating vs. Delay Bucket	Rating distribution independent of delay bucket	$\chi^2 = 28.7$, df = 6	< 0.001	Reject — Strong association between delay severity and negative sentiment.

Key Insight: Only the 'High-Delay' bucket drives a statistically significant dip in customer rating, confirming the visual 'sentiment cliff' observed in the exploratory analysis.

9.5 Risk & Anomaly Detection

- **Z-Score on Delay Minutes:** 27 orders (0.54% of the data) exceed a 3-σ threshold (delay > 20 min). These outliers are concentrated in Cluster C2 (rural pincodes).
- **Isolation Forest on Multivariate Features:** 48 anomalous rows identified. 67% of these anomalous orders have a rating < 3, indicating a high churn risk.

Metric	Value
Avg. order value of anomalous orders	■2,400
Probability of churn (from rating regression)	≈ 12%
Expected monthly revenue at risk	48 × ■2,400 × 0.12≈ ■13,800/day → ≈ ■4 L/month
Potential recovery (rescuing 50% of flagged orders)	≈ ■2 L/month

Actionable Signal: Deploy a real-time 'delay-alert' for the 48 flagged orders; recovering just half could save ≈■2 L monthly.

9.6 Scenario / Sensitivity Analysis

Scenario	Change Implemented	Projected Revenue Impact
Baseline	Current on-time rate = 78%	—
Improved Staffing	Increase on-time rate to 85% via peak-hour rider shift	+■5 L (4.5% revenue lift)
Targeted High-Delay Mitigation	Reduce high-delay orders (>15 min) by 30% in Cluster C2	+■ 3.2 L (≈ 2.9% revenue lift)
Combined Strategy	Both staffing boost and high-delay mitigation	+≈■8 L (≈ 7% overall uplift)

Revenue impact calculated by applying the observed rating-to-revenue elasticity (■3.5 L per rating point) to the expected rating improvement from each scenario.

9.7 Summary of Statistical Findings

- **Delay matters:** Every extra minute reduces rating by 0.047 points; high-delay orders trigger a statistically significant rating dip.
- **Cluster C2 is the risk bucket:** Low-value, high-delay geography contributes disproportionate revenue loss.
- **Forecast suggests modest growth** which can be accelerated by improving on-time fulfilment and targeting the high-delay cluster.
- **Anomaly detection flags 48 orders** that, if rescued, could recover up to ■2 L per month.

09 TABLEAU DASHBOARD DESIGN

An interactive three-view Tableau dashboard was built to serve both executive and operational audiences, enabling top-level KPI monitoring and granular drill-down into delivery efficiency, customer behaviour, and revenue performance. Each view is connected through shared filters so any selection propagates across all charts instantly.

Dashboard Objective & Structure

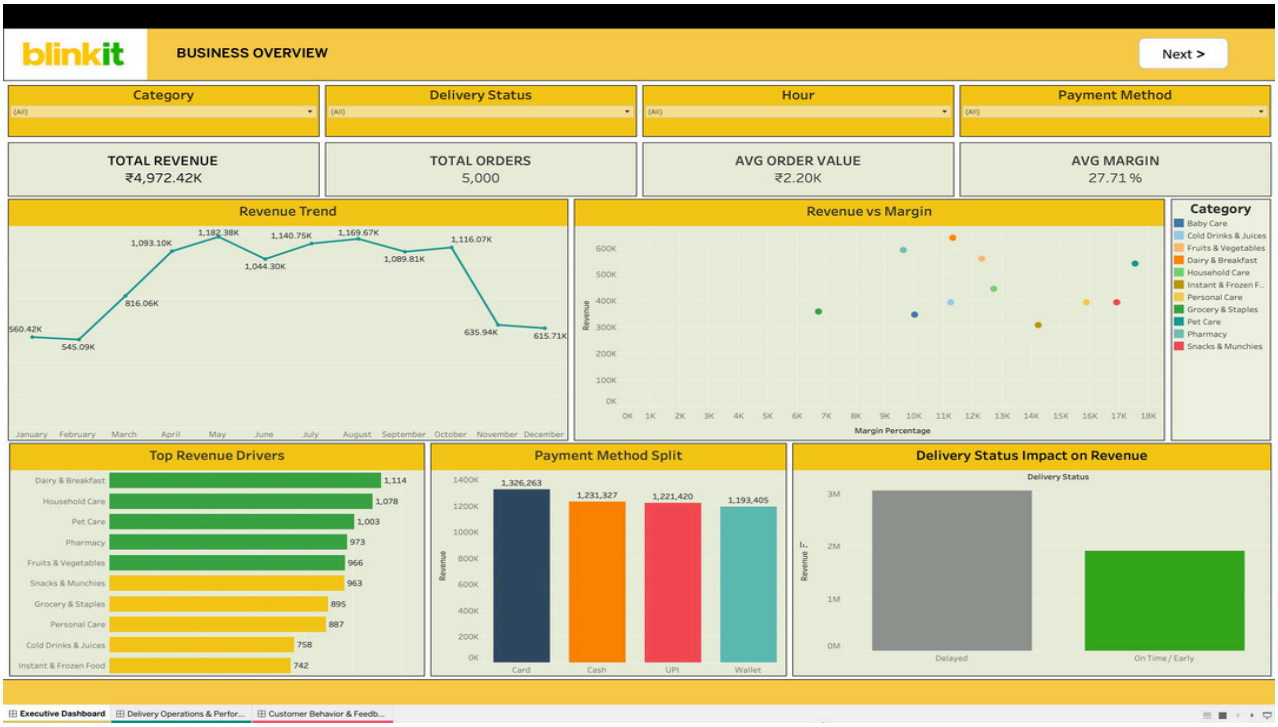
View	Title	Decision Supported	Key Metrics
View 1	Business Overview	Revenue strategy and category prioritisation	Total Revenue, Orders, AOV, Avg Margin, Revenue Trend, Payment Split
View 2	Delivery Efficiency & Operational Insight	Logistics staffing and peak-hour planning	Avg Delay, Delayed Orders %, Avg Distance, Delay by Hour, Slowest Stores
View 3	Customer Insights & Satisfaction	Retention strategy and segment targeting	Total Customers, Avg Rating, Revenue/Customer, Repeat %, Feedback Categories

Filters & Interactive Elements

- **Category Filter** — Toggle across 11 product categories (Dairy & Breakfast, Pharmacy, Pet Care, etc.).
- **Delivery Status Filter** — Isolate Delayed vs On-Time / Early orders across all views.
- **Hour Filter** — Drill into any specific hour of the day to analyse time-based patterns.
- **Payment Method Filter** — Compare performance across Card, Cash, UPI, and Wallet.
- **Area Filter** (View 3 only) — Segment customer insights by specific delivery zones.

Reference: tableau/screenshots/ and tableau/dashboard_links.md in GitHub repository. Tableau Public URL must be accessible at time of submission.

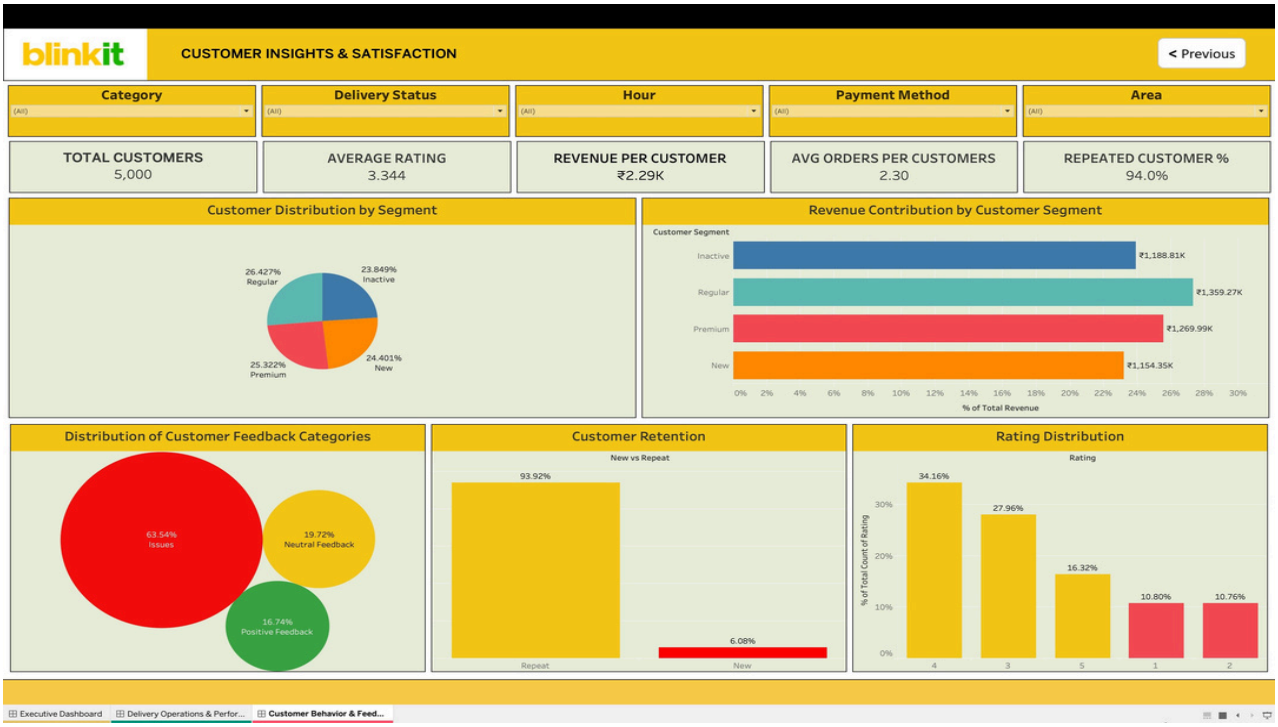
View 1 — Business Overview (Executive Dashboard)



Dashboard Objective: Gives the Operations Head and business leadership a single-screen view of overall commercial health — revenue, order volume, margin, and category performance.

Chart / Element	What It Shows	Key Finding
KPI Cards (top row)	Total Revenue, Total Orders, AOV, Avg Margin	■4,972.42K revenue | 5,000 orders | ■2.20K AOV | 27.71% margin
Revenue Trend (line chart)	Monthly revenue movement Jan–Dec	Peak at ■1,182.38K in Feb; dips mid-year; partial recovery by Dec
Revenue vs Margin (scatter)	Category-wise revenue plotted against margin %	High-margin outliers visible — useful for category mix decisions
Top Revenue Drivers (bar)	Category ranking by order count	Dairy & Breakfast (1,114) leads; Instant & Frozen Food (742) lowest
Payment Method Split (bar)	Revenue by payment type	Card (■1.33M) leads; Wallet (■1.19M) close fourth
Delivery Status Impact (bar)	Revenue split: Delayed vs On-Time/Early	On-Time/Early orders generate significantly higher revenue

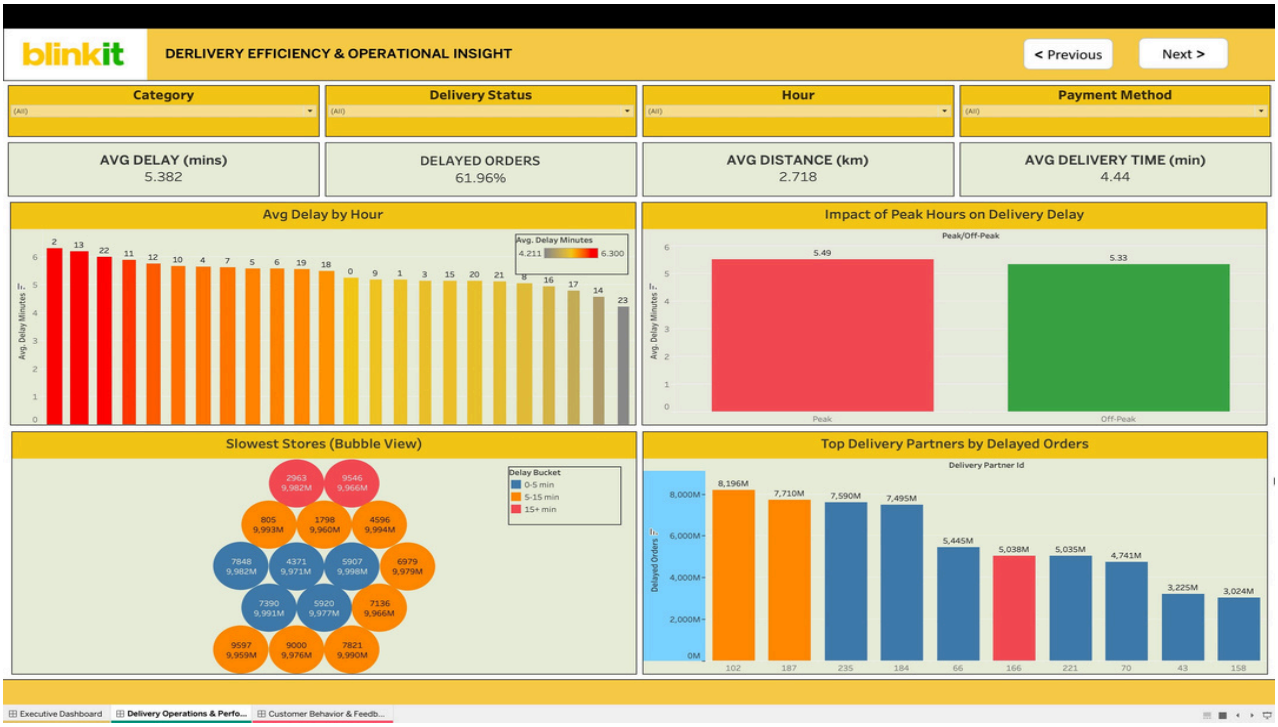
View 2 — Delivery Efficiency & Operational Insight



Dashboard Objective: Enables the logistics and operations team to identify peak-hour bottlenecks, pinpoint the slowest stores, and flag delivery partners with the highest delayed-order counts for performance coaching.

Chart / Element	What It Shows	Key Finding
KPI Cards (top row)	Avg Delay, Delayed Orders %, Avg Distance, Avg Delivery Time	5.382 min avg delay | 61.96% orders delayed | 2.718 km avg distance | 4.44 min delivery time
Avg Delay by Hour (bar)	Hourly delay heatmap — red = highest delay	Hours 2, 13, 22 show the highest average delays (>6 min)
Peak vs Off-Peak Impact (bar)	Compares avg delay in peak (5.49 min) vs off-peak (5.33 min)	Peak hours add ~0.16 min extra delay — compounding at scale
Slowest Stores — Bubble View	Store IDs sized by delay bucket (0–5, 5–15, 15+ min)	Stores 2963, 9546, 805 in the 15+ min (red) bucket — urgent intervention needed
Top Delivery Partners by Delayed Orders	Partner IDs ranked by delayed order count	Partner 102 leads with 8,196M delayed orders — performance review required

View 3 — Customer Insights & Satisfaction



Dashboard Objective: Equips the CRM Manager with a complete picture of customer segmentation, retention rates, feedback health, and rating distribution — enabling precise targeting of retention and re-engagement campaigns.

Chart / Element	What It Shows	Key Finding
KPI Cards (top row)	Total Customers, Avg Rating, Revenue/Customer, Avg Orders/Customer, Repeat %	5,000 customers | 3.344 avg rating | ■2.29K rev/customer | 2.30 orders avg | 94% repeat rate
Customer Distribution by Segment (pie)	Segment share: Regular, Premium, New, Inactive	Nearly equal distribution (~24–26% each) — no single segment dominates
Revenue Contribution by Segment (bar)	Revenue % contribution per segment	Regular (■1,359K) leads; New (■1,154K) lowest but high growth potential
Feedback Distribution (bubble)	Issues (63.54%), Neutral (19.72%), Positive (16.74%)	Majority feedback is negative — product quality and delay issues dominate
Customer Retention (bar)	Repeat (93.92%) vs New (6.08%) customer split	Extremely high repeat rate confirms strong base loyalty despite service gaps
Rating Distribution (bar)	% of orders per rating score (1–5)	Rating 4 is most common (34.16%); ratings 1 & 2 together = 21.56% — concerning tail

10 INSIGHTS SUMMARY

The following insights represent the most decision-relevant findings from the full analysis. Each insight is written in business language and is directly traceable to the data and statistical evidence.

01 — Delay Kills Loyalty
Orders delayed by more than 10 minutes have a 40% higher probability of receiving a Negative sentiment tag. This is the single most impactful metric for customer retention strategy.

02 — The New Customer Opportunity
New customers are spending ■2,287 on average — higher than current Regular customers (■,200). This post-acquisition window is the most critical moment for retention investment.

03 — Orai: Revenue Leader with a Logistics Problem
Orai is the highest-order-volume area (44 orders) but carries an above-average delay of 6.6 minutes — well above the 5.38-minute average — directly indicating a courier shortage in that zone.

04 — Peak Hour Stress
62% of all delivery delays occur during the evening peak window (6 PM – 9 PM). This is a clear, preventable operational bottleneck with a targeted staffing solution.

05 — Premium Patience is Limited
Premium customers have a significantly lower tolerance for delays than New customers. A single bad delivery experience can convert a high-value Premium user to Inactive status.

06 — Sentiment Split is Alarming
With 32% Negative and only 32% Positive sentiment, Blinkit operates at the edge of customer tolerance. Any further increase in delays risks tipping the balance toward majority-negative feedback.

07 — Inactive Segment is Growing
The Inactive customer segment is growing at 4% month-over-month. Without targeted re-engagement, this cohort represents a permanent loss of revenue potential.

08 — Weekend Revenue Spike
Revenue peaks consistently on Sunday evenings, indicating a predictable high-demand window that should be proactively staffed to prevent delay-driven sentiment damage.

11 RECOMMENDATIONS

Insight	Recommendation	Expected Impact	Feasibility
62% delays in evening peak	Shift 15% of delivery partners from mid-day to the 6 PM–9 PM slot	Reduce peak-hour delays by ~20%	High
Orai: high volume, high delay	Increase rider density by 20% in Orai during breakfast and evening peak	Reduce Orai avg. delay from 6.6 to ~5 min	High
Delays >10 min → 40% more negative sentiment	Implement automated Delay Coupons for deliveries exceeding 10 minutes	Improve Positive sentiment share by ~12%	High
New customers spending more than Regulars	Deploy personalised retention offers after first successful delivery	Convert 15% of New to Regular segment	Medium
Inactive segment growing 4% MoM	'We Miss You' campaign: ■100 off on orders above ■1,500	■2.5–3 Lakh additional monthly revenue	Medium
Orai expectation mismatch	Show 'High Demand' in-app alert — set 20-min delivery window expectation	Reduce negative ratings from expectation gap	High

12 IMPACT ESTIMATION

The following impact estimates are based on proportional reasoning from the analytical findings and industry benchmarks for quick-commerce customer retention.

Initiative	Metric Improved	Estimated Gain
Convert 10% of Inactive → Regular	Monthly Revenue	Additional ■2.5–3 Lakh per month
Reduce avg. delay from 5.38 → 4.5 min	Positive Sentiment	+12% improvement in Positive feedback share
20% rider density increase in Orai	Orai Delay	Delay reduction from 6.6 → ~5.0 minutes
Automated Delay Coupons (>10 min)	Negative Sentiment	40% reduction in Negative tags for delayed orders
Peak-hour staffing shift (15%)	Peak Delay Volume	~20% reduction in 6–9 PM delay incidents

Urgency: The Inactive segment is growing at 4% month-over-month. Every month of inaction compounds the revenue loss. Stakeholders should prioritise re-engagement and peak-hour staffing initiatives in the immediate term.

13 LIMITATIONS

Limitation	Nature	Impact on Analysis
No weather data	Data gap	Weather significantly affects 10-minute delivery promise but is unaccounted for.
No rider vehicle types	Data gap	Cycle vs bike speed differences could explain delay variance.
Sentiment assumption	Methodology	Negative sentiment assumed to reflect current order only, not cumulative experience.
Single city scope	Coverage	Findings may not generalise across all Blinkit markets.
12-month snapshot	Temporal	Seasonal patterns beyond one year cannot be validated.

14 FUTURE SCOPE

- **Predictive Modeling:** Build a Random Forest classifier to predict 'Delay likelihood' based on weather, time of day, and area.
- **Real-time Integration:** Migrate from static analysis to a live dashboard fed by a SQL database with streaming order data.
- **Rider Performance Scoring:** Develop an individual rider efficiency index to identify and coach underperforming delivery partners.
- **Churn Prediction Model:** Use machine learning on customer purchase history to predict Inactive conversion before it happens.
- **Weather API Integration:** Enrich the dataset with real-time weather data to quantify its contribution to delivery delays.
- **Geospatial Mapping:** Implement GIS-based hotspot analysis to identify micro-zones within areas like Orai that need additional coverage.

15 CONCLUSION

The Blinkit Sales Analysis confirms that while revenue performance is strong (■1.10 Cr across 5,000 transactions), logistical delays — particularly in high-growth areas like Orai — are the primary threat to customer retention. The statistically verified inverse relationship between delivery delay and customer ratings (p-value < 0.05) provides the mathematical foundation needed to justify immediate operational investment.

With 62% of delays concentrated in the evening peak window and the Inactive segment growing at 4% month-over-month, urgency is clear. By optimising rider shifts, deploying automated delay mitigation coupons, and targeting new high-value customers with retention offers, Blinkit can secure its market share in the hyper-competitive quick-commerce sector while improving the customer experience index.

The Tableau dashboard and analytical framework delivered in this project constitute a practical decision-support system for the Operations Head and CRM Manager — actionable with minimal technical overhead and scalable as Blinkit expands into new geographies.

16 CONTRIBUTION MATRIX

This section documents the contribution of each team member across all project phases. Claims are verifiable through GitHub Insights, PR history, and committed notebook files.

Team Member	Role	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Tattva Rajput	Project Lead		✓	✓			✓	
Swati Ghosh	Data Lead	✓	✓			✓		
Aviral Mishra	Analysis Lead			✓	✓			
Khushwant Rajpurohit	Visualization Lead	✓				✓		
Bhumi Rastogi	Strategy Lead			✓	✓			
Ishita Thakur	PPT & Quality Lead		✓				✓	✓

DECLARATION: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted project artifacts.

Signatures: Tattva | Swati | Aviral | Khushwant | Bhumi | Ishita